

Section 3: Exploratory Data Analysis (EDA)

Report

Mobile Money Transaction Analysis — User Classification

1. Dataset Overview

Metric	Value
Total transactions	20,530
Unique users	69 (51 original + 18 anonymized)
Date range	2023-03-10 to 2026-03-28
Operators	MobileMoney (96.3%), OrangeMoney (3.7%)
Features	24 columns (13 base + 11 engineered)
Missing values	New_balance: 57 (0.3%), days_since_last_txn: 69 (0.3%)

The dataset was produced by the Section 2 cleaning pipeline from 51 original raw SMS export files and 18 pre-collected anonymized files. All transactions are classified into 6 types with 0 unclassified records.

2. Descriptive Statistics

Numerical Variables

Variable	Mean	Median	Std Dev	Skewness
Amount (XAF)	7,454	500	33,702	12.7
New_balance (XAF)	20,362	2,653	59,199	6.4
net_amount (XAF)	-4,148	-300	30,038	-8.7
days_since_last_txn	0.67	0.08	3.55	13.8
monthly_txn_frequency	35.5	22.8	54.2	5.1
send_receive_ratio	4.62	3.24	5.79	2.6
txn_velocity_7d	10.3	5.0	14.9	3.9

Key observation: All financial variables exhibit strong positive skewness, indicating the dataset is dominated by small, frequent transactions with a long tail of large-value operations.

Categorical Distributions

Transaction Type	Count	%
transfert	6,592	32.1%
transaction	5,104	24.9%
airtime	3,502	17.1%
paiement	2,637	12.8%

Transaction Type		Count	%
retrait		2,487	12.1%
depot		208	1.0%
Direction		Count	%
OUT		15,920	77.5%
IN		4,610	22.5%

3. Visualizations Summary

11 publication-quality visualizations were generated, exceeding the minimum requirement of 8:

#	Visualization	Type	File
1	Distribution of Key Financial Variables	Histogram + KDE	01_amount_distributions.png
2	Box Plots of Transaction Amounts	Box plot	02_amount_boxplots.png
3	Monthly Transaction Volume Over Time	Time series (bar + line)	03_monthly_time_series.png
4	Hourly Transaction Distribution	Bar + area chart	04_hourly_distribution.png
5	Correlation Heatmap	Heatmap	05_correlation_heatmap.png
6	Categorical Variable Distributions	Bar charts (2x2)	06_categorical_barcharts.png
7	User-Level Relationship Exploration	Scatter plots	07_scatter_relationships.png
8	Grouped Comparisons	Grouped bar charts (2x2)	08_grouped_comparisons.png
9	Activity Heatmap (Hour x Day)	Advanced: Heatmap	09_activity_heatmap.png
10	User Classification Distribution	Bar + histogram + violin	10_user_classification.png
11	Class Balance & Per-Class Profile	Bar + grouped bar	11_class_balance_check.png

All visualizations include proper titles, axis labels, legends, and formatted tick marks.

4. Key Insights (5 Patterns Identified)

Insight 1: Predominantly Micro-Transactions with Extreme Skewness

The median transaction amount (500 XAF ~ \$0.80 USD) is 15x smaller than the mean (7,454 XAF). With skewness > 12, the dataset is dominated by small airtime purchases and transfers, while a small number of large deposits and payments create the long tail. **Implication:** Transaction frequency is a more stable classification feature than raw amount.

Insight 2: Spending-Dominant Behavior (77.5% Outbound)

Over three-quarters of all transactions are outgoing, driven by transfers (32%), bundle purchases (25%), and airtime (17%). Deposits account for only 1% of transactions. **Implication:** The send/receive ratio captures fundamental behavioral differences between net-senders and net-receivers.

Insight 3: Strong Evening Peak (17:00-20:00) Across All Days

The activity heatmap reveals consistent peak transaction volume during post-work evening hours (17:00-20:00), with a secondary morning peak (09:00-12:00). Weekend activity accounts for ~29% of

total volume. **Implication:** Temporal features (hour, weekday, is_weekend) carry meaningful behavioral signal.

Insight 4: Distinct User Populations Between Data Sources

Anonymized users (18 users) transact with 25x higher mean amounts (110,932 XAF vs 4,431 XAF) but at lower frequency. Original users (51 users) show high-frequency, low-value patterns typical of personal mobile money use. **Modeling safeguard:** The `data_source` feature will be **excluded** from the classification model to ensure the classifier learns behavioral patterns rather than memorizing dataset identity. This prevents a shortcut that would not generalize to new users.

Insight 5: Transaction Velocity Strongly Separates Activity Classes

High-activity users show dramatically higher 7-day rolling transaction velocity and monthly frequency compared to Low-activity users. These temporal activity metrics directly align with the classification target. **Note:** Because these features overlap with the target variable (total transaction count), they require careful handling during modeling to avoid data leakage (see Section 5 below).

5. User Classification Definition

Methodology

Users were classified into three behavioral segments using **quantile-based thresholds** on total transaction count (per-user aggregation):

Class	Threshold	Users	Avg Txn Count	Avg Monthly Freq
Low	<= 33rd percentile (~55 txns)	~23	~22	~8
Medium	34th-66th percentile (~55-204 txns)	~23	~118	~24
High	> 66th percentile (>204 txns)	~23	~753	~83

Why 3 Classes (Not 2 or 4)?

- **Business relevance:** Financial service providers typically segment users into casual, regular, and power tiers for marketing, credit scoring, and product design
- **Statistical balance:** Tercile-based splits produce approximately equal class sizes (~23 users each), avoiding class imbalance that would penalize minority-class prediction
- **2 classes** would lose the distinction between occasional and regular users; **4+ classes** would create groups too small (< 18 users) for reliable modeling with only 69 users

Class Balance Verification

The class balance was explicitly verified (see Visualization 11):

- All three classes contain approximately equal numbers of users

- The per-class feature profiles show clear separation: High-activity users have higher monthly frequency but lower mean transaction amounts, while Low-activity users show higher individual transaction values but infrequent use

Data Leakage Awareness

Important: The classification target is based on total transaction count. Features like `monthly_txn_frequency` and `txn_velocity_7d` are derived from the same underlying quantity. In Section 4 (Modeling), this will be addressed by either:

1. Using a **temporal split** — train on historical data, predict future activity class
2. **Excluding direct leakage features** and using only behavioral profile features (`mean_amount`, `send_receive_ratio`, `pct_weekend`, etc.) that describe *how* a user transacts, not *how much*

6. Connection to Prediction Goal & Feature Recommendations

Prediction Target

User Activity Classification: Predict whether a user falls into the Low, Medium, or High activity category based on their transaction history and behavioral features.

Recommended Features (by priority, with leakage assessment)

Priority	Feature	Rationale	Leakage Risk
High	<code>mean_amount / median_amount</code>	Transaction size profile	None
High	<code>send_receive_ratio</code>	Behavioral pattern indicator	None
High	<code>avg_balance (New_balance)</code>	Financial capacity proxy	None
Medium	<code>pct_weekend</code>	Lifestyle/usage pattern	None
Medium	<code>std_amount</code>	Transaction variability	None
Caution	<code>monthly_txn_frequency</code>	Activity level (overlaps target)	High — temporal split only
Caution	<code>txn_velocity_7d</code>	Recent intensity (overlaps target)	High — temporal split only
Exclude	<code>data_source</code>	Proxy for dataset identity	Shortcut — always exclude
Low	<code>Operator</code>	Limited signal (96% MobileMoney)	None

Hypotheses to Test in Modeling

1. Behavioral features (`mean_amount`, `send_receive_ratio`, `pct_weekend`) can classify users without frequency-based leakage features
2. Mean transaction amount alone is insufficient — it needs combination with other behavioral features
3. Send/receive ratio separates agents/merchants from personal users

7. EDA Limitations

1. **No demographic data in current analysis.** The assessment requires demographic/contextual data (age, profession, zone, income) collected via questionnaire. Once Section 1 questionnaire responses are available, additional analyses should include activity by profession, zone, income range, and smartphone ownership.
2. **Skewed financial data.** All financial variables are heavily right-skewed (skewness > 10 for Amount). Log-transformations or robust scaling should be applied before modeling.
3. **Imbalanced data source representation.** Original data (51 users, 19,261 txns) vastly outweighs anonymized data (18 users, 1,269 txns). Per-user aggregation mitigates this, but transaction-level analyses are dominated by original-dataset patterns.
4. **Temporal coverage varies by user.** Some users have 3 months of data while others have up to 36 months. Features like `monthly_txn_frequency` partially normalize this, but users with short histories may have less stable profiles.
5. **Data leakage potential.** The classification target (total transaction count) overlaps with frequency-based features. Section 4 must address this through temporal splits or feature exclusion.